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HERITAGE RAILWAY ASSOCIATION

GUIDANCE NOTE

STEAM LOCOMOTIVE BOILERS - Introduction to Guidance Documentation & Glossary

Purpose

This document describes good practice in relation to its subject to be followed by Heritage Railways, Tramways and similar bodies to whom this document applies.

Endorsement

This document has been developed with and is fully endorsed by Her Majesty's Railway Inspectorate (HMRI), a directorate of the Office of Rail Regulation (ORR).

Disclaimer

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Supply

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Users of these Guidance Notes should check the HRA website to ensure that they have the latest version.

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1. Introduction

a) This Guidance Note is the introductory document to a series of guidance notes dealing with locomotive boilers and associated components that were produced and are updated as required by the “Steam Locomotive Boiler Codes of Practice” practitioners’ meetings.

b) This series of Guidance Notes has been developed from documentation generated over the last hundred years in many countries which was reviewed by a panel of UK experts under the guidance of HRA and HMRI. It represents the current view on Good Practice in the Industry and as such is subject to change without notice. The latest version of this document and any subsidiary documents will be available on the Heritage Railway Association website www.hra.gb.com

c) This series of documents is focussed on the requirements of main-line, standard gauge and narrow gauge railway locomotive boilers typically running on 15” (381mm) or greater gauges. It may also be helpful for operators of stationary steam engines with similar boilers, traction engines and miniature steam locomotives

2. Recommendations

a) This guidance note is issued as recommendations to duty holders.

b) Where duty holders decide to take actions that are not in agreement with these recommendations following appropriate risk assessments or for other reasons, it is recommended that those decisions are reviewed by the senior management body of the organisation concerned and a formal minute is recorded of both the decision reached and the reasons for reaching it.

3. General

a) Railway locomotive boilers are designed to create, store and distribute steam at high pressure. The working life of such a boiler can be considerably shortened if due care is not taken at all stages of inspection, repair, running maintenance and day-to-day running.

b) In the past there have been a series of accidents and explosions due to work being undertaken without having due regard to the inherent risks involved. It is with that in mind that HMRI and HRA set up the series of meetings of boiler practitioners to discuss the issues; distil good practice and codify it into this series of Guidance Notes.

c) This guidance is written for the assistance of people competent to perform these tasks. In places the terminology used may be specific to such practitioners.

d) This guidance should also be useful to those in a supervisory or more general role. However no work should be undertaken unless the persons concerned are deemed competent to do so.

4. Scope of this Series of Documents

a) The initial aim was to create the definitive Guidance Notes on Boiler Repair and Reconstruction. It was quickly realised that two other strands were needed; one for general, legal and background information and the other for those areas where operating procedures and techniques can affect safety or seriously shorten the life of boilers.

b) The work therefore divides into three areas:

   i) Repair and reconstruction.

   ii) Operations related.

   iii) General and Legal.

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c) The series covers all aspects of the boiler including those identified on the diagram below

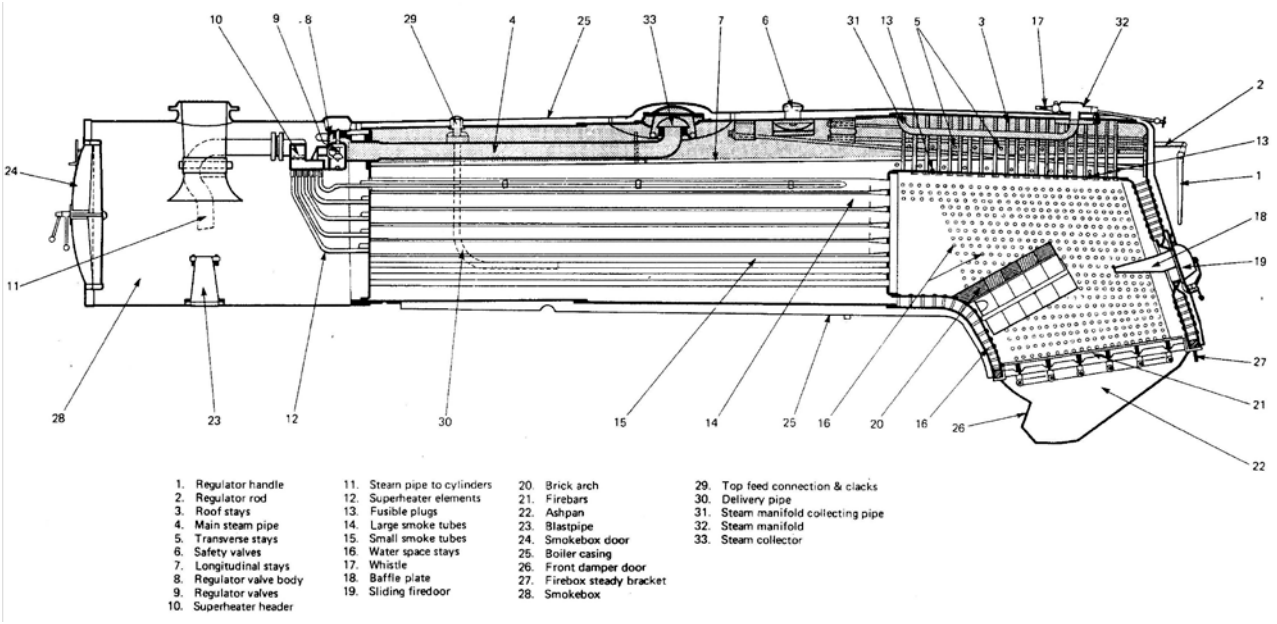


Diagram 1- Typical Large Superheated Boiler

5. Dimensional Notation

a) The dimensions in this document are variously described in a mixture of imperial and metric units. Where practical, equivalent dimensions have been shown but in some cases the dimensions do not easily equate and so the units in force at the time that the original designs were documented have been used.

6. Terminology

a) A Glossary has been provided in an Appendix to cover the main specialist terms used as well as definitions of some common terms to explain their particular application.

7. Documentation Control

a) The issued documents are available in the Guidance section of the HRA website: www.heritagerailways.com

b) The top of each page of each document gives the reference number, Issue number and date of issue.

c) The issue number is increased when corrections or changes need to be made.

d) Each document is reviewed at intervals of five years or earlier should changes need to be made more urgently.

e) Users are encouraged to advise HRA of any suggested changes or comments on existing Guidance Notes by e-mailing those comments to boiler.comments@hra.gb.com with the document number in the title line.

f) A log is maintained of requests for alterations in the interval between issues which is regularly reviewed with the lead authors of each document.

8. Issue Status

a) Those with no issue number or date in black are not available at present.

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9. Series Framework

a) This document is the key introductory document.

HGR-B9000	Steam Locomotive Boilers	Iss 01	November 2014

b) It includes an Introduction to Guidance Documentation, Glossary & Reference Documentation

10. Repair and Reconstruction Guidance Notes

a) This section provides guidance on all aspects of boiler repair

b) The documents in this section are in the series HGR-B90xx

HGR-B9001	Tubing of Locomotive Boilers	Iss 02	November 2014
HGR-B9002	Boiler Stays	Iss 01	April 2012
HGR-B9003	Superheaters	Iss 01	April 2012
HGR-B9007	Boiler Mountings, Pipework and Fittings	Iss 01	April 2012
HGR-B9008	Fusible Plugs	Iss 02	November 2014
HGR-B9009	Washout Plugs	Iss 01	August 2008
HGR-B9011	Safety Valves	Iss 01	April 2012
HGR-B9014	Mudhole Doors	Iss 03	January 2011
HGR-B9020	Platework	Iss 01	April 2012
HGR-B9021	Inner Firebox Repair and Renewal	Iss 01	April 2012
HGR-B9022	Thermic Syphons & Arch Tubes	Iss 01	October 2014
HGR-B9023	Steel Rivets and Riveting	Iss 01	April 2012
HGR-B9024	Welding Procedures & Processes (applicable to Loco Boilers)	Iss 01	October 2014
HGR-B9025	Patch Screws	Iss 01	June 2012
HGR-B9030	Smokeboxes	Iss 01	June 2012
HGR-B9050	Steam Heating Apparatus (for Locomotives and Carriages)	Iss 01	October 2014

11. Operations related Guidance Notes

a) This section provides guidance only on those aspects of operations that affect the safety and life of the boiler.

b) These operations related guidance notes are not about “how to drive” but very much about making sure that safety is assured in the short term and long term due to effect of driving and firing techniques.

c) The documents in this section are in the series HGR-B91xx

HGR-B9118	Boiler Water Treatment	Iss 02	June 2012
HGR-B9140	Preparation, Operation & Disposal	Iss 01	November 2014
HGR-B9155	Washout and Examination	Iss 01	November 2014
HGR-B9160	Examination in service	Iss 01	November 2014
HGR-B9170	Frost and Corrosion Protection	Iss 01	November 2014

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12. General Legal and Reference Guidance Notes

a) This section provides general background information.

b) The documents in this section are in the series HGR-B92xx

HGR-B9210	Legal Background (Incl PSSR & PED) + RSP6	Under development
HGR-B9220	Materials and Non-Destructive Testing	Iss 01 October 2014
HGR-B9230	Quality Assurance	Iss 01 November 2014
HGR-B9260	Written Scheme of examination & prep. for Annual Examination & Competent Person	Under development

Note: When HGR-B9210 and HGR-B9260 are available this document will be updated

13. Acknowledgements

a) The project was initiated by ORR's HMRI and the HRA to document best practice. We owe a significant debt to the HMRI for providing the initial drive for the whole of this project.

b) Most of the personnel actively engaged in Locomotive Boiler Construction and Repair in the U.K. were involved in the creation of this series of documents and gave comments and/or engaged in discussions at the many co-ordinating meetings. Some only made a few crucial comments. Others, to whom we owe a very great debt, were deeply involved in the creation of the framework and the many individual Guidance Notes. Some, alas, are no longer with us but all contributed their expertise in order that the Steam Locomotive Boiler maintenance process could be documented and made available to future practitioners and their managers. Without all these contributions we would have been unable to create the documentation.

end of main document

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Appendix A – Glossary of Terms

Scope

This glossary covers items and the specialised process associated with the document suite forming the Boiler Code of Practice. It does not cover all other general engineering processes that would otherwise make this document exhaustive.

Terminology listed below in **RED** should not be used as it is either colloquial, possibly misleading or a non-recommended alterative.

Specific meaning of verbs used	**Definition**
may	where the guidance suggests options
must / shall	only used where there is a legal requirement for the measures described to be employed. Where applicable a reference to the relevant act or regulations will be provided
should	the primary verb for statements of guidance which in most circumstances would be expected to be carried out
Boiler Code of Practice terminology	**Definition or Description of item/ process**
Accumulation Test	A performance test to prove safety valves ability to prevent excessive over pressurisation of a boiler producing steam approaching its maximum steaming rate.
Approved Welder	A suitably qualified and experienced welder who has undertaken and passed the appropriate welding qualification and procedure tests for a particular welded joint design.
Anneal	Process of softening (by heating and cooling).
Arch Tube	Steel tubes fitted in the firebox and connected to the lower portion of the firebox tube plate at their lower end and to points near the top of the firebox door plate at their upper end.
Back head	The plate at the back of the firebox on which the regulator, gauges etc are mounted; see also - Door plate - outer
Blowdown valve	A valve fitted to the lowest part of the boiler to facilitate the removal of sediment
Boiler tube hole bush	A thin metal bush designed to be fitted into an oversized tube plate hole. This is intended to reduce the tube hole diameter back to a size that allows standard // original sized tubes to be refitted.
Boiler tube Ferrule	Ferrules at the firebox end of boiler tubes.
Boiler Tubes - fire tubes	A group of tubes typically between 1 and 2 inches in diameter in the boiler barrel connecting the firebox to the smoke box. A proportion of heat from the fire is transferred via the tube wall to the water surrounding them.
Boiler tube beading	The beading of the ends of boiler tubes to the tube plate.
Boiler tube seal welding	The welding of the ends of boiler tubes to the tube plate.
Boiler tube swaging	A process that reduces the boiler tube diameter at one end to suit the tube plate hole size.

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Boiler tube swelling	A process that increases the boiler tube outside diameter (normally the smoke box end) to suit the tube plate hole size
Boxing up	Closing and sealing up of a boiler; refitting all mountings and plugs
Bridge (dog)	Fastening clamp used to draw the mud hole door onto its seat
Butt strap	A steel plate riveted to and joining the ends of the butt joint of a longitudinal seam.
Bulging (see also; Quilting)	Bulging is due to broken stays where the boiler pressure has pushed the firebox plate towards the fireside in an area that extends beyond some stays. (see also Quilting)
Caulking	Closing the edge of a rivet or seam with a purpose made tool in order to make the joint leak proof. (see also Fullering)
Change	Remove the original, and fit new or overhauled part or assembly in its place
Check	Determine a particular nominated condition before, during or after repair, e.g. completeness, security, position, corrosion, etc.
Coded Welder	An American system of welder capability approval; see Approved Welder
competent person	A person competent and qualified to undertake the task described
Competent Person	A competent individual person or a competent body of persons corporate or unincorporated; as referenced in the Pressure Systems Safety Regulations (2000).
Crinoline Bands	Framework to support boiler cladding
Crown sheet	The upper sheet of the inner firebox.
Defect / Defective	Any fault or faults in a component or assembly, which may prevent the component or assembly from fulfilling its designed purpose, e.g. cracking
Deflector	A removable semi-circular plate normally retained in position above the fire hole intended to deflect secondary air into the fire stream and away from the tube ends.
Dezincification	Corrosion of an alloy containing zinc (usually brass) involving loss of zinc
Dismantle	Take to pieces
Dome	A steam space set as high up as possible arranged to house the regulator/steam collector and auxiliary steam pipes where the steam is at its driest condition.
Door Plate - inner	The rear inner firebox plate containing the fire door ring or forged flanging
Door Plate - outer	The rear outer flanged plate of a boiler
Examiner	A person qualified in the appropriate trades necessary to undertake examination of a locomotive prior to it entering service. The tasks are not to be confused with duties of Operating Personnel during preparation of motive power for service.
Expansion bracket	'L' shaped steel sections fitted to each side of a firebox. This supports the weight of the boiler at the firebox end. The lower section is arranged to rest on the locomotive frame, the arrangement having clearance in order to allow sliding due to boiler expansion when hot.
Firebox ring	See Foundation Ring
Foundation Ring	A steel ring shaped to correspond with the perimeter of the lower extremity of a firebox, rounded at the corners and of a thickness that separates the inside and outside sheets of the firebox to which it is attached.

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Fullering	As caulking except that a broader tool is used to act on the full plate thickness of a riveted seam. (See also Caulking)
Fusible plug	A warning device in the form of a screwed plug inserted into the crown of a firebox. having a core (normally of pure lead) – designed to melt in the event of firebox crown becoming uncovered due to low water level.
Gag	This is a type of clamp device used to prevent a safety valve from lifting. It is only used during the safety valve setting operation by trained and authorized maintenance personnel. Where more than one safety valve needs to be set individually it is convenient to restrain the other valves from lifting for the purpose of its adjustment.
Gauge	Determine a nominated dimension by using suitable measuring equipment, e.g. ruler, micrometer, calipers, feeler gauge, or Go/NoGo gauge.
Girder bars	See Stays - Girder
Grooving	Fatigue corrosion acting in certain areas where cyclic stresses are applied
Heat Treatment	A combination of heating and cooling operations applied to a metal or alloy to obtain desired metallurgical conditions or properties
Inspect	Determine conformity to required standards during and after overhaul and repair.
Inverted Tee link stays	See Stays - Sling
Knuckle radius	A tight simple or compound radius in a flanged plate.
Lacing	See Patch Screws
Lagging	Material with insulating properties applied to hot surfaces to retain heat
Lap Joint	A method of securing the ends of boiler plate or of two plates laying the edge of one over the other then riveting them together.
Lead plug	See Fusible plug
Mud hole door;(MHD)	Doors in the firebox provided to assist the removal of deposits.
NDT	Non-Destructive Testing
Normalise	A heat-treatment process that has the object of relieving internal stresses and refining the grain structure
Patch screws	Special countersunk screw designed to screw into a specially prepared threaded and countersunk rivet hole in order to replace worn out rivets where a seam cannot be re-riveted
Pipework	A pipe or system of pipes together with associated valves, pumps, compressors and other pressure containing components and includes hose or bellows. This description does not include any protective devices
Pitting	Localised severe corrosion
Pressure System	A system comprising on or more Pressure Vessels of rigid construction, any associated Pipework and Protective Devices
Progressive threads	A method of redeeming thread condition by re-cutting an existing threaded hole slightly larger (normally 1/16 inch increment) in order to recover a damaged or worn thread form.
Protective Devices	Devices designed to protect the system against system failure and devices designed to give warning that system failure might occur
Quilting	Quilting is usually due to overheating and / or thinned plates where boiler pressure has distorted the firebox plate towards the fireside in an area between a number of stays (See also Bulging)

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Reassemble	Put together
Record / Records	Put down in writing / enter in a computer system, the result of any specified examination, test or inspection, in accordance with defined procedures.
Refit	Put back and reconnect. The term "Refit" is to be used as the term "Replace" can lead to confusion in terms of work carried out.
Relevant Fluid	Steam at any pressure and/or Compressed Air with a pressure at or above 0.5 bar should also be considered a relevant fluid (See PSSR2000)
Remove	Disconnect and take off
Renew	To replace an item with a new or overhauled item.
Replace	See Refit
Responsible Person	The individual appointed by that part of a railway organisation responsible for the use of a boiler. This person will have responsibility to co-ordinate all the necessary work and to ensure the safety of the boilers and pressure vessels on the railway
Rivet	A fastener installed through multiple plates where the shank is formed into a head
Roof stays	See Stays - Crown
Safe operating limits	The pressure limits beyond which the system should not be taken.
Safety Critical	A system or component whose malfunction or failure may result in injury or death
Safety Valves (Various)	A mechanism which automatically releases steam from a boiler when the pressure exceeds pre-set limits.
Scale	A layer of foreign material adhered to a surface, e.g. corrosion scale or deposits of boiler water contaminants
Scrap	Dispose of in a safe manner which prevents its re-use.
Set screws	See Patch Screws
Smoke tubes ‘smalls’	See Boiler Tubes – fire tubes
Stays - Girder	A longitudinal or transverse girder which has sufficient stiffness to support the fire box crown.
Stays - Sling	A special design of vertical stays which allows upward movement of the firebox crown adjacent to the tubeplate when raising steam.
Stays - Longitudinal	A supporting stay bar directly connected between the front tube plate and boiler back head.
Stays - Crown	Generic term for ordinary screwed stays which are screwed between the firebox roof and the boiler outer wrapper sheet.
Stays - Palm	A steel bar riveted to the boiler barrel which connects to the firebox below the tube bank in order to support the area of a firebox tube plate.
Stays - Flexible	Stays that allow a degree of flexibility of movement. They incorporate a spherical head that fits into a corresponding cup socket in the outer wrapper sheet to facilitate flexibility
Stepped studs, oversized, incremental threads	See progressive threads
Strip	Take off covering, e.g. paint, polish, fabric
Stud	A threaded fastener
Stud Driver	A device used when fitting studs

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Superheater	A device used to convert saturated steam or wet steam into dry steam.
Super heater tube-Bottle end	A tube adaptor section that connects the super heater flue tube to the firebox. Depending on design, this can be either;- - a copper swaged tube brazed to the flue tube and expanded into the firebox, or – a thick walled steel swaged tube welded to the flue and screwed into the firebox tube hole.
System Failure	Unintentional release of stored energy or the scalding effect of steam from a pressure system
Test	Prove correct operation by specified trial
Test Cock	Supplementary boiler water level indicator.
Thermic syphon	A funnel-like metal assembly within the firebox that connects the low point of the boiler with the top part of the boiler.
Trial Cock	See Test Cock
Throatplate	Front plate of the outer firebox casing which joins the firebox to the barrel
Tubeplate radius	See Knuckle radius
User	In relation to a pressure system means the employer or self employed person who has control of the operation of the pressure system
Waist sheet stay	See Stays - Palm
Washout	The process of cleaning scale, sludge and sediment from the water spaces of the boiler
Water leg	See Water space
Water space	The part of the boiler between the firebox and outer wrapper that is filled with water. Designated as side, front or back water spaces / legs.
Water Treatment	A process used to make water more acceptable for use in a boiler
Welt	See Butt strap
Working Pressure	The maximum steam pressure that the boiler is allowed to operate at. This pressure is set by the design of the boiler. It is protected from over pressurization by safety valves set at this pressure.
Written Scheme of Examination	The written scheme for the periodic examination by a competent person, of the following parts of the system: All protective devices Every pressure vessel in which a defect may give rise to danger Those parts of the Pipework in which a defect may give rise to danger

This glossary has been developed by the group with additional items added from the Rail Industry Safety and Standards Board of Australia document:

“Inspection, maintenance and repair of rail locomotive boilers– Code of Practice” RISSB-v1-01 June 2011

end of appendix

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Appendix B – Selected references on the Care & Maintenance of Locomotive Boilers

	A	B	C	D	E	F	H
1	**Title**	**Author**	**Paper**	**Journal**	**Source**	**Year**	**Comment on contents**
2	**Reference Books**						
3	A Manual of Steam Locomotive Restoration and Preservation (Hardback book)	Harvey D. W.			D & C	1980	
4	Handbook for Railway Steam Locomotive Enginemen (Hardback Book)	Harvey R.F. ?			B.T.C.	1957	Standard BR reference book
5	Locomotive Engineers' Pocket Book				Locomotive Publishing Company		
6	Steam Locomotive Design: Data and Formulae	E.A. Phillipson			Locomotiv	1936	Cambden reprint 2004
7	Industrial Water Treatment	Hamer, Jackson & Thurston			Butterwor	1961	
8	Locomotive Management -Cleaning, Driving Maintenance	Lee C. E.odgson & Lake			St Margar	1948	General locomotive care - One chapter on the Boiler. (editions 8, 9 or10) (Hodgson & Lake)
9	Maintenance of Locomotive Boilers (Hardback book)	Garraway A			Ian Allan	1999	
10	Motive Power Organization and Practice (Hardback official)	Rudgard H.	53894		B.T.C.	1946	Contains details of 'X' scheme of repair (p38)
11	Locomotive Boiler Explosions	Hewson C H			D & C	2003	
12	Raising Steam on the L.M.S. - The Evolution of LMS Locomotive Boilers (Hardback book)	Cook A. F.			RCTS	1999	
13	**Published papers**						
14	Tube Failures	Groom S.W.	72	39	I.Loc.E.	1919	
15	Engine Failures	Clayton J.	232	85	I.Loc.E.	1928	
16	The Taper Boiler	Thompson J.W	361	134	I.Loc.E.	1936	
17	Locomotive Feed Water Treatment	Hancock J. S.	366	136	I.Loc.E.	1937	
18	The Locomotive Boiler	Turner J.	375	138	I.Loc.E.	1937	
19	Copper and Copper Alloys for Locomotive Firebox Construction	Cook M.	393	146	I.Loc.E.	1938	
20	Economical Locomotive Running Shed Operation	Pargiter G.M.	397	147	I.Loc.E.	1939	
21	A treatment of Locomotive Feed Water	Pinchen D. B.	404	149	I.Loc.E.	1938	
22	The Solution to Some After Problems of Water Softening	Topham W.L.	410	152	I.Loc.E.	1939	
23	Modern Locomotive Boiler Shop Practice	Fell E.W.	414	154	I.Loc.E.	1940	
24	Corrosion of Boiler Tubes	Turner T.H.	438	170	I.Loc.E.	1942	Chief Chemist LNER. How tubes corrode; prevention of corrosion. Paper also in I.Mech. E. Volume 149 of 1943.
25	The Work of their Craft (Presidents Address)	Whalley F.S		193	I.Loc.E.	1946	
26	A Brief History of the Application of Base Exchange Water Softeners on Railways (South America)	Walter A.J.	463	195	I.Loc.E.	1947	

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	A	B	C	D	E	F	H
1	**Title**	**Author**	**Paper**	**Journal**	**Source**	**Year**	**Comment on contents**
27	Organisation and Carrying Out Exams. & Repairs of Locos at Running Sheds in Relation to Loco Pref. & Availability	Rudgard H.	464	196	I.Loc.E.	1947	L.M.S. 'X' scheme details, Cooling (9hrs), washing-out (2 hrs), filling & raising steam (8hrs) for cl. 5-7 boilers
28	Water Softening for Locomotive Boilers	Hancock J. S.	468	198	I.Loc.E.	1947	L.M.S. Hardness levels at which you should soften as well as treat water.
29	Some Developments in Locomotive Workshop Practice 1939-1948	Forsyth I.C.	485	209	I.Loc.E.	1949	
30	The Influence of the Treatment of Boiler Waters on the Maintenance and Utilisation of Steam Locomotives.	Armand. L		210	I.Loc.E.	1949	This is the reference paper on the SNCF TIA 'In Tank Water Treatment System'.
31	The Early History and Later Application and Development of Superheating in Locomotive Practice	York R. S.		211	I.Loc.E.	1949	The paper was presented in New South Wales. Tube and element faults.
32	The Application of Welding to Locomotive Boiler Copper Fireboxes	Harrison J. F.	511	227	I.Loc.E.	1952	Mechanical & Electrical Engineer BR (LM) Derby. Discussion includes K.J.Cook, R.C. Bond
33	The Design and Construction of Steel Fireboxes	Compton J. N.	525	234	I.Loc.E.	1953	Yorkshire Engine Company. Identifies stress points.
34	A Brief History of Locomotive Feed Water Treatment on the L.M. Region of British Railways	Hancock J. S.	573	257	I.Loc.E.	1957	Water Treatment Assist. to CM & EE BR(LM) Derby. Common discussion with paper 572. + Bibliography by TH Turner.
35	Some Aspects of Locomotive Boiler Feedwater Treatment	Parsons A. J.	572	257	I.Loc.E.	1957	Technical Officer, Railway Section I.C.I Alkali Division (Alfloc Water Treatment) In depth paper
36	Experience with the Steel Fireboxes of the Southern Region Pacific Locomotives	Burrows M. G. Wallace A. L.	584	262	I.Loc.E.	1958	15 years of operating experience with MN & BoB boilers. Modifications justified.
37	Steel for Railway Purposes	Dearden & Roberts	586	263	I.Loc.E.	1958	
38	Corrosion of Boiler Tubes	Turner T.H.		149	I.Mech.E	1943	Copy of I.Loc E. J170 Vol 32
39	Presidential Address	Bulleid O.V.S.		156	I.Mech.E	1947	A review paper - Photos of Stays and Boilers
40	Some Notes on the 'Merchant Navy' Class Locomotives of the Southern Railway	Bulleid O.V.S.			I.Mech.E	1946	Design, construction and early operation. Novel features including welding and assembly of fireboxes covered.
41	Fusion-Welded Boiler on D & H Locomotive (USA)	Edmonds G.S.		Journal	A.S.M.E.	1946	
42	Locomotive Boilers - Welded Construction (USA)	Partington J		Journal	A.S.M.E.	1946	
43	Experience with Intercrystalline Cracking on Railroads (USA)	Bardwell & Laudemann		Transaction	A.S.M.E.	1940	
44	Heat Transfer in the Locomotive Boiler (USA)	Fry L. H.		Transaction	A.S.M.E.	1946	
45	Boilers, The Heart of the Matter - Part 1 Certification	Anderson J	1	USA	L&RP	1986	Practical loco boiler maintenance

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	A	B	C	D	E	F	H
1	**Title**	**Author**	**Paper**	**Journal**	**Source**	**Year**	**Comment on contents**
46	Boilers, The Heart of the Matter - Part 2 - The Firebox	Anderson J	2	USA	L&RP	1987	Practical loco boiler maintenance
47	Boilers, The Heart of the Matter - Part 3 - Boiler Inspection	Anderson J	3	USA	L&RP	1987	Practical loco boiler maintenance
48	Boilers, The Heart of the Matter - Part 4 - Tools	Anderson J	4	USA	L&RP	1987	Practical loco boiler maintenance
49	Boilers, The Heart of the Matter - Part 5 - Tube & Flue Removal	Anderson J	5	USA	L&RP	1987	Practical loco boiler maintenance
50	Boilers, The Heart of the Matter - Part 6 - Preparing & Installing Tubes	Anderson J	6	USA	L&RP	1988	Practical loco boiler maintenance
51	Boilers, The Heart of the Matter - Part 7 - Staybolts	Anderson J	7	USA	L&RP	1988	Practical loco boiler maintenance
52	Boilers, The Heart of the Matter - Part 8 - Keeping Valves Tight	Anderson J	8	USA	L&RP	1988	Practical loco boiler maintenance
53	Boilers, The Heart of the Matter - Part 9 - Throttle & Special Valves	Anderson J	9	USA	L&RP	1989	Practical loco boiler maintenance
54	Boiler Corrosion and Water Treatment B.R. 1335	Admiralty	1335		HMSO	1945	Types of Corrosion with illustrations
55	Modern water treatment and Sodium Aluminate for Railways	Beal & Stevens	179	ndon Eng	GWR	1930	Basic Discussion
56	**Notes of Current Practice**						
57	Apexior Paint Trials	Pickering P.			Swanage	2002	Controlled trial results
58	Boiler Water Treatment	Heintzman M			SVR	2001	Overview of Current Best Practice
59	Locomotive Boiler Maintenance, Tubes and Retubing	Snell J.B.			RH&DR	2000	Review of RH&DR boiler issues.
60	Steel Boiler Materials and Use (paper for Bulleid Pacific Locomotive Association)	Johns M.A.			WSR	2002	Past BR (S) Boiler Water Expert
61	SVR Water Treatment & Corrosion References	Heintzman M			SVR	2001	
62	Feed Water Treatment for Road Engine Boilers	Garnett R.J.			N.T.E.T.		Water Treatment Basics for Traction Engine Owners

end of appendices

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